

- 3 Fig. 3.1 shows lines joining points at the same potential in and around a pair of parallel, charged metal plates. The plates have a separation of 2.5 cm.

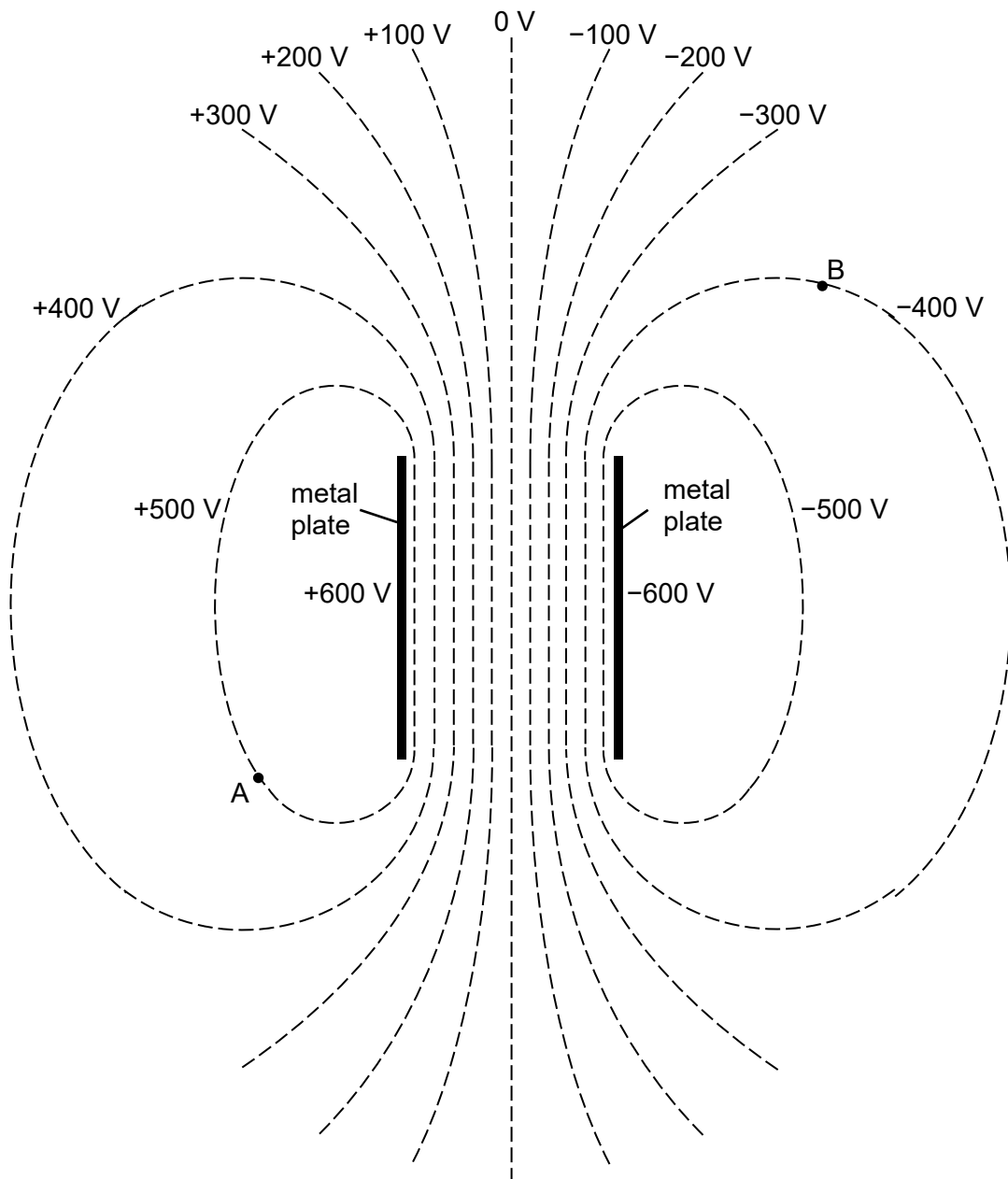


Fig. 3.1

- (a) (i) State the relation between electric field strength E and potential V .

.....
..... [1]

- (ii) By reference to Fig. 3.1, suggest why the electric field between the plates is uniform.

.....
..... [1]

(iii) Determine the value of the electric field strength between the plates.

electric field strength = NC^{-1} [2]

(b) On Fig. 3.1, draw six lines to represent the electric field between the plates. [2]

(c) A particle moves from point A to point B. As a result, there is a loss in potential but the particle gains electric potential energy of $1.44 \times 10^{-16} \text{ J}$.

(i) Calculate the charge of the particle.

charge = C [2]

(ii) Suggest a particle that might have this value of charge.

..... [1]

(iii) Hence, calculate the speed of the particle at B, given that its initial speed at A is $3.5 \times 10^7 \text{ m s}^{-1}$.

speed = m s^{-1} [2]

1. (a) A resistor of resistance of $9.4 \text{ k}\Omega$ and a LDR (light dependent resistor) are connected